

// Exercise 7: Financial Forecasting

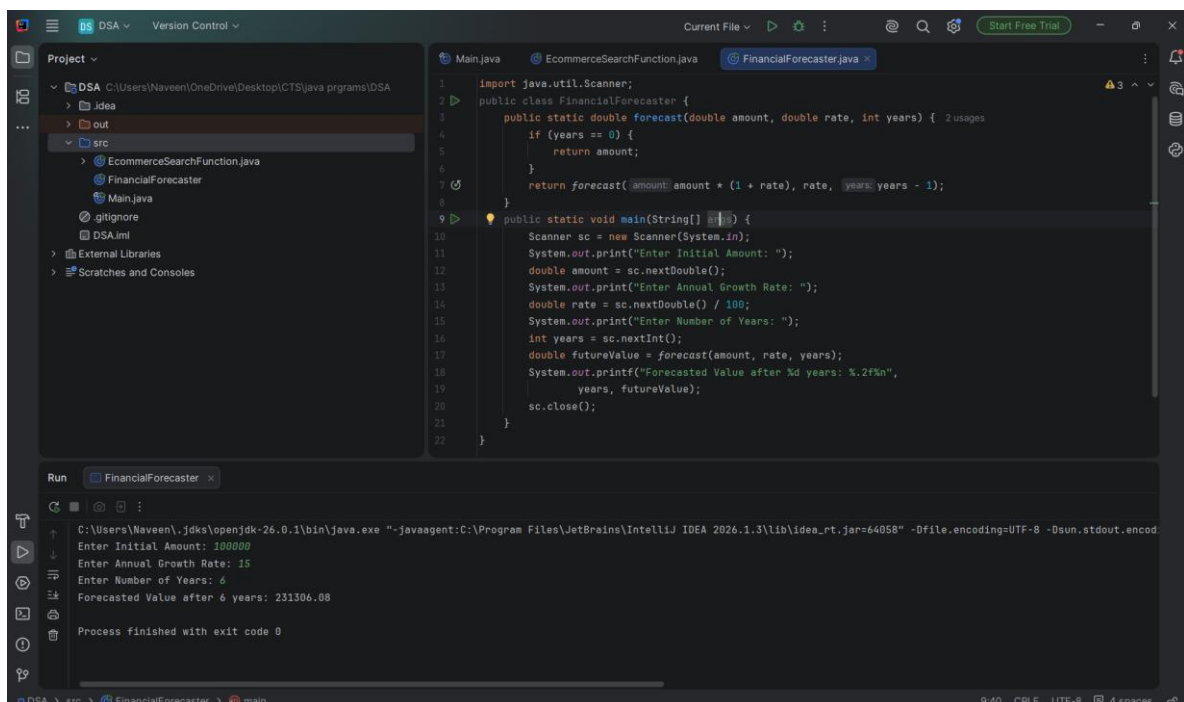
Program:

```
import java.util.Scanner;

public class FinancialForecaster {
    public static double forecast(double amount, double rate, int years) {
        if (years == 0) {
            return amount;
        }
        return forecast(amount * (1 + rate), rate, years - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Initial Amount: ");
        double amount = sc.nextDouble();
        System.out.print("Enter Annual Growth Rate: ");
        double rate = sc.nextDouble() / 100;
        System.out.print("Enter Number of Years: ");
        int years = sc.nextInt();
        double futureValue = forecast(amount, rate, years);
        System.out.printf("Forecasted Value after %d years: %.2f%n",
            years, futureValue);
        sc.close();
    }
}
```

Output:



The screenshot displays an IDE window with the following components:

- Project View:** Shows a project named 'DSA' with a source folder 'src' containing files like 'EcommerceSearchFunction.java', 'FinancialForecaster', 'Main.java', and 'DSA.iml'.
- Editor:** Displays the code for 'FinancialForecaster.java', which matches the code provided in the previous block. It includes a 'forecast' method and a 'main' method that uses a 'Scanner' to take user input.
- Run Console:** Shows the execution of the program. The output is as follows:
Enter Initial Amount: 100000
Enter Annual Growth Rate: 15
Enter Number of Years: 6
Forecasted Value after 6 years: 231306.08
Process finished with exit code 0